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# Jamming avoidance trajectory planning and load balancing user association in mmWave UAV-assisted HetECN☆

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ABSTRACT

Emergency communication network (ECN) can provide fast, efficient and high-capacity communication services for specific areas by using mmWave transmission and unmanned aerial vehicles (UAVs) serving as aerial base stations (ABSs) or relay nodes. Now, in order to satisfy diverse demands, ECN should support different types of nodes, access methods, and traffic distributions, which is referred to as heterogeneous ECN (HetECN). Therefore, inappropriate trajectory planning and unbalanced traffic loading can lead to UAV flight collisions and network congestion. In this article, we jointly optimize UAV jamming avoidance trajectory and user association strategy aimed to load balancing, to maximize the utilization of HetECN. Specifically, an improved artificial potential field (APF) method along with mmWave beam forming technology is used to obtain the jamming avoidance trajectory of UAVs, and the optimal deployment location of UAVs are determined based on the distribution of ground users (GUs). Subsequently, the matching game and alliance game are comprehensively used to determine the load balancing based GU-UAV associated strategy under various GU demands, thereby ensuring traffic load balancing and resource optimization allocation. In addition, altitude fine-tuning have been made to further power consumption, thereby improving overall network efficiency. Simulation results demonstrate that the proposed method can achieve the expected performance in network utilities such as coverage rate, network capacity, load balancing effect of mmWave UAV-assisted HetECNs.

## 1. Introduction

In the past few decades, emergencies events caused by various natural disasters, such as floods and earthquakes, as well as military operations, have generated to enormous losses to human life and ecology. In order to control the development of emergency events, it is particularly crucial to establish an efficient emergency communication network (ECN) in time [1]. However, due to the damage of road traffic facilities and the threat of further emergencies, it is difficult for rescue personnel and emergency vehicle (EV) to directly enter the disaster area. Obviously, additional support is needed to achieve effective connections in ECNs.

Unmanned Aerial Vehicle (UAV)-assisted network, utilizing UAV as airborne base stations (ABSs) or relay nodes, has been proposed to solve these problems [2]. Due to characteristics such as low latency, cost, and insensitivity to environmental influences, UAVs now play an important role in emergency communication support, disaster relief, post-disaster reconstruction and military reconnaissance [3]. The main applications of UAVs can be classified as ubiquitous coverage, auxiliary relay, and data collection [4,5]. Firstly, UAVs can enhance network coverage of ECN by assisting ground base stations (GBSs) in providing seamless coverage anywhere within the service area. ECN is usually heterogeneous [6], yet the heterogeneity of network architecture and communication technology poses numerous challenges to the quality of service (QoS) in heterogeneous ECN (HetECN). In this situation, UAVs are utilized to help restore necessary communication services in HetECN where ground communication infrastructures are mostly damaged [7]. The second application of UAV is relay, which refers to providing communication connections to GUs without reliable communication links in ECN [8]. Finally, UAVs can also make data collection. Compared to transmitting data to GBS at high power consumption, sensors can transmit data to UAVs flying near them to have low-energy communication connections, thereby extending the lifespan of them in HetECNs [9]. To sum up, if deployed properly, UAVs can enhance the capabilities of HetECN for disaster prediction, damage assessment, and rescue operations. However, for emergency scenarios with numerous obstacles and inevitable interference, it is necessary to design appropriate trajectory of UAVs based on special characteristics to ensure their arrival at deployment locations safely.

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After the hovering positions of UAVs are determined, the division of service areas between UAVs has a profound impact on the overall utility of HetECN, which becomes another important issue needs to be solved [10]. Firstly, the division of service areas affects the channel gain between GU and its serving UAV. If the serving area is too broad, GUs at the edge of it will suffer severe path loss [11]. Secondly, the division of service areas also determines the number of GUs served by each UAV. As the number of GUs increases, the bandwidth of each GU decreases due to limited total bandwidth of UAV [12]. In this case, GUs in sub regions with high traffic loads of UAV will have low throughput, resulting in unbearable latency for emergency communication [13]. Besides, in actual HetECN, the heterogeneity of GUs results in their various traffic demands, leading to the division of UAV service areas should not only considering the number of GUs, but also depending on the different traffic demands of them [14]. Therefore, in order to achieve optimal utilization of network resources, how to arrange the association between UAV and GU based on load balancing is a crucial question needs to be solved.

At the same time, in the current field of wireless communications, millimeter-wave (mmWave) technology, with its extremely high bandwidth and abundant spectrum resources, has become a critical support technology for 5G and future 6G networks. The application of mmWave technology in HetECN scenarios has become an irreversible trend due to its large bandwidth and low latency [15]. However, owing to its sensitivity to obstacles and other characteristics, the performance evaluation of the combination of mmWave technology and HetECN is worthy of in-depth research [16].

Given this context, based on the requirements of emergency communication tasks and complex external environmental characteristics in mmWave UAV-assisted HetECN, this paper proposes a novel joint optimization framework to study how to achieve UAV jamming avoidance trajectory and load balancing GU-UAV association in HetECN. Firstly, we use the improved APF method for UAV trajectory design, utilizing multiple repulsive forces and mmWave antenna to ensure interference management and collision avoidance. Then, by sequential using the match and alliance game theory, we develop a GU-ABS association strategy to balance the maximum traffic demands between UAVs. Finally, a UAV altitude fine-tuning method is proposed to reduce the overall power consumption of HetECN.

The main contributions of this paper can be summarized as follows:

- A UAV-assisted communication system supported by emergency vehicle (EV) is designed to solve the problem of rapid communication recovery in HetECN, auxiliary UAVs equipped with mmWave antenna are connected with EVs. Experimental results have shown that this is an efficient and feasible HetECN model that can be used as a reference.
- This paper proposes an improved APF algorithm to solve the problem of jamming avoidance UAV trajectory planning in HetECN. Comprehensively taking the constraints of complex obstacles, mission area boundary, GU hotspots and malicious jamming into account, finally obtain the optimal trajectories and hovering positions of UAVs in HetECN.
- We innovatively propose a distributed GU-UAV association strategy for the load balancing problem of UAVs based on game theory. By sequentially using matching game and alliance game, the total network traffic demand is expressed as the logarithm sum of various traffic demands of GUs, balancing the heterogeneity of HetECN and the fairness of UAV traffic allocation.
- A novel method integrating improved APF method and game theory is used to solve the utility optimization problem of HetECN, which comprehensively considering network heterogeneity and environment complexity, meeting the high requirements in network latency, traffic, etc. Experimental results show that our proposed strategy significantly improves the overall network utilization.

The rest of this paper is organized as follows. Section 2 introduces the current methods about the UAV-assisted HetECN. In Section 3, network model and its connectivity definition are introduced. The proposed algorithms are introduced in Section 4. Simulation results are presented in Section 5. Finally, Section 6 gives some concluding remarks.

## 2. Related works

In recent years, mmWave UAV-assisted network has received widespread attention, especially in executing tasks such as natural disaster rescue and battlefield cooperative operations, in which multiple UAV trajectory design [3,17–29] and traffic load balancing [30–41] are two essential aspects for successful task execution and network efficiency improvement. In addition, this article also introduces APF method, the game theory and their applications [22,40,42–45].

Actually, the UAV related technologies such as deployment density [17], hovering position [18] and trajectory optimization [46] have got rapid development in recent years. Facing with the complex external environment, numerous obstacles, and high demand in UAV-assisted HetECN, UAV trajectory design is an important and fundamental aspects that is widely concerned by researchers. Prevalent methods for UAV path planning can be roughly divided into five categories, and their respective advantages and disadvantages are summarized as follows. Firstly, the Reinforce learning (RL) based methods are able to self adjust strategies when the environment changes, but often with high training costs, and the strategies they learned are always difficult to explain and verify [21]. Secondly, The node based methods have a large amount of mature algorithms and theoretical support, such as Dijkstra and A-star (A*) algorithms [22,23], but when the environment or constraint conditions change frequently, it may have to recalculate the path, and the computational complexity may be high for large-scale networks. Thirdly, methods based on mathematical models, they can provide precise UAV optimization trajectory [24], but the results highly depend on the accuracy of the mathematical model, when facing real-world influencing factors outside the model, their flexibility and adaptability are poor. Fourthly, methods based on improved heuristic algorithms, such as genetic algorithm (GA) and gray wolf optimization (GWO) [25,26], they have high computational efficiency in solving complex and large-scale problems, but may not guarantee finding the optimal solution, and the quality of these solutions highly depends on parameter selection. The fifth is fusion methods, they often combine multiple methods to overcome their respective drawbacks and enhance path planning performance [27,28]. Combining these methods with ECN, numerous scholars have achieved advancements in satisfying the energy consumption and power control [3,29]. However, The deployment of HetECN requires comprehensive consideration of low latency requirements, high obstacle avoidance ability of ABS, network heterogeneity, and short-term network traffic surgery, The APF method has the characteristics of high efficiency, dynamic adaptability and flexibility, making it a valuable attempt to solve UAV trajectory planning in HetECN.

The load balancing problem has garnered extensive attention in HetECN. It addresses congestion and high latency issues in hotspot areas by offloading some traffic to neighboring underloaded cells [30]. Existing researches on load balancing are widely applied in Internet of Things (IoT), wireless sensor network (WSN), and edge computing [31]. Abundant works have been conducted on the load balancing problem in traditional terrestrial networks with different objectives, such as system throughput [32,33], Energy Efficiency (EE) [34,35], Spectrum Efficiency (SE) [36], and multi-objective optimization [37]. However, for mmWave UAV-assisted ECN, additional constraints such as high timeliness and large short-term traffic flows need to be considered. From the perspective of research methods, most current studies solving UAV load balancing through convex optimization and heuristic algorithm. Ref. [38] proposes an effective Swap-and-Judge Simulated

**Table 1**
List of main notations.

| Notation | Description |
|---|---|
| $\mathcal{R}$ | Ground mission area |
| $\mathcal{M}, \mathcal{N}$ | The set of ABSs and GUs |
| $\varpi_u^m, \varpi_v^k, \varpi_g^n$ | Location information of ABS, GBS and GU |
| $\varpi_a, \varpi_v$ | Location information of AEJ and EV |
| $\mathbf{P}_{\mathrm{LoS}}, \mathbf{P}_{\mathrm{NLoS}}$ | Probability of LoS and NLoS channel |
| $\mathcal{L}_{LoS}^{m,n}, \mathcal{L}_{NLoS}^{m,n}$ | LoS/NLoS path loss |
| $\overline{\mathcal{L}}_m(x, y)$ | Average path loss |
| $a, b$ | Constants related to environment |
| $\mathfrak{D}_{\mathrm{a}}$ | The gain of ABS directional antenna |
| $p_G, p_A$ | The transmission power of EV and ABS |
| $R_{m,n}$ | Achievable rate for $\mathrm{GU}_n$ associated with $\mathrm{ABS}_m$ |
| $\mathcal{A}_{m,n}$ | The GU-ABS association index |
| $L_m$ | The length of trajectory of $\mathrm{ABS}_m$ |
| $\mathcal{E}_m$ | The trajectory of $\mathrm{ABS}_m$ |
| $R_{total}$ | Total rate requirement of HetECN |
| $\mathcal{U}\left(R_{total}\right)$ | Total utility function of HetECN |
| $\tilde{\mathcal{U}}_n^m, \tilde{\mathcal{U}}_m^n$ | The estimated utility function of ABS and GU |
| $\succ_{\mathcal{M}}, \succ_{\mathcal{N}}$ | The preference relationship of ABS and GU |
| $\Phi^*$ | The optimal matching results |
| $\Theta^t$ | Set of GU-ABS association |
| $\theta_m$ | The GUs set served by $\mathrm{ABS}_m$ |
| $T_m^{total}$ | The flight time of $\mathrm{ABS}_m$ |
| $h_{opt}$ | The optimal altitude of ABS |
| $K_{att}, K_{rep}$ | Coefficient constant of attraction/repulsive field |
| $\Theta^{final}, \Theta^{ini}$ | GU-ABS initial and final association group |

Annealing (SJSA) algorithm for task allocation and UAV path planning, which balances the tasks between UAVs and achieves satisfactory timeliness in scenarios such as natural disaster rescue, and battlefield collaborative operations. Ref. [39] proposes a heuristic algorithm to solve the mixed-integer programming problem, jointly optimizing trajectories to maximize UAV energy utilization efficiency and minimize communication delay. Actually, most studies tend to deploying more UAVs in areas with high GU density [41], which do not take the varying traffic demands of GUs in actual HetECN into consideration. Moreover, few studies on achieving HetECN load balancing consider how to safely navigate UAVs to mission areas. Thus, researches combining trajectory design and traffic load balancing in HetECN is still lacking.

The APF method is a classic obstacle avoidance method for UAVs, in which UAV moves according to the combined force of repulsion and gravity, in order to reach the target while avoiding obstacles. Due to its simple mathematical principles, low computational complexity, and minimal resource consumption, the APF method has been widely applied and improved [46], especially for UAV movement control [40]. On the other hand, the basic elements of the game theory include players, strategy sets, and utility. Each player aims to comprehensively consider the interactions between all players and find strategies to maximize their utility [42,43]. In previous researches, game theory has been widely applied to wireless resource allocation, such as power control [44], spectrum sharing [45] and routing selection [22], to obtain more efficient and optimized UAV deployment solutions, implementing HetECN load balancing through game theory seems to be an efficient and achievable approach. What is more, UAVs equipped with large bandwidth and high-speed mmWave antenna can better achieve interference isolation, which has been confirmed by our previous research [47]. In addition, using UAVs equipped with intelligent surface as airborne RIS (ARIS) can reduce interference attacks and enhance legitimate transmission [48,49], and unmanned ground vehicle carrying backup batteries moves along with UAVs can also compensate for the shortage of UAV energy [50]. but how to apply these methods to UAV-assisted HetECN is still a topic worth deep exploration (see Table 1).

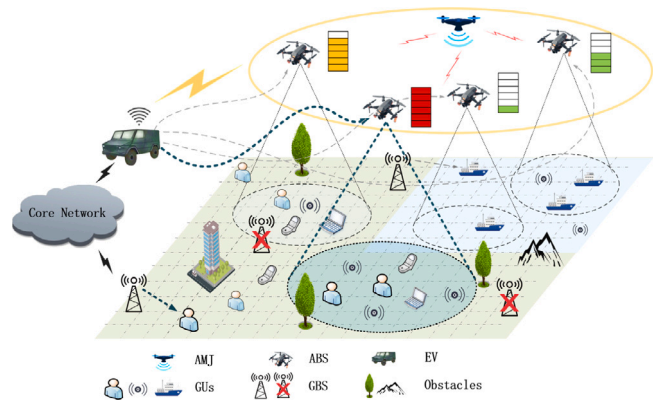



**Fig. 1.** The scenario of mmWave UAV-assisted HetECN.

## 3. System model

### 3.1. mmWave UAV-assisted HetECN

Fig. 1 illustrates a scenario of mmWave UAV-assisted HetECN under jamming conditions, a given rectangular ground task area $\mathcal{R}$ is divided into several non-overlapping cells, the center of each cell can accommodate one GU. GBSs are mostly damaged and unable to support full coverage of the mission area. Due to the complex terrain environment, the EV cannot drive in that area, $M$ ABSs equipped with mmWave antenna are developed to assist a small number of available GBSs. After receiving task, they take off from the EV to provide communication services for $N$ heterogeneous GUs. According to GU's various traffic requirements, their traffic load situation is displayed on their right side. Due to the presence of obstacles and malicious jamming in HetECN, it is necessary to design executive jamming avoidance trajectories of ABSs to avoid them. The EV acts as a relay connecting the core network and ABSs, its signal can achieve full coverage of ABSs in the mission area. ABSs are equipped with mmWave antenna array to enhance signal strength in specific directions and isolate interference.

Assuming that the EV is deployed at $\varpi_e = \left(x_e, y_e, 0\right)$ on the edge of the task area, and 3D coordinate of the $M$th ABSs are $\varpi_u^m = \left\{x_u^m, y_u^m, h_u^m\right\}$, $m \in \{1, 2, \ldots, \mathcal{M}\}$, the GBSs that can provide services are located in $\varpi_v^k = \left\{x_v^t, y_v^t, h_v^t\right\}$, $v \in \{1, 2, \ldots, \mathcal{V}\}$. If a GU is within the service range of GBS, GBS will be prioritized as the service base station, and the remaining GUs will be provided services by ABSs. The center coordinates of heterogeneous GUs $\varpi_g^n = \left\{x_g^n, y_g^n, 0\right\}$, $n \in \{1, 2, \ldots, \mathcal{N}\}$ are covered by at least one ABS, it indicates that the GU is within the cover range. Furthermore, we assume that the aerial external jamming (AEJ) is located at $\varpi_a = \left\{x_a, y_a, h_a\right\}$, significantly impacting trajectories of ABSs.

Due to the complex external environment of HetECN, the transmission links are composed of both Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) situations. The Low Altitude Platform (LAPs) path loss model proposed in [51] is used to represent the A2G large-scale fading, which can be represented as

$$\mathcal{L}_{\mathrm{LoS}}^{m,n} = 20\log\left(\frac{4\pi f_c d\left(\varpi_u^m, \varpi_g^n\right)}{c}\right) + \eta_{\mathrm{LoS}} \tag{1}$$

$$\mathcal{L}_{\mathrm{NLoS}}^{m,n} = 20\log\left(\frac{4\pi f_c d\left(\varpi_u^m, \varpi_g^n\right)}{c}\right) + \eta_{\mathrm{NLoS}} \tag{2}$$

where $f_c$ is the carrier frequency, $c$ represents the speed of light, $d\left(\varpi_u^m, \varpi_g^n\right) = \sqrt{\left(x_g^n - x_u^m\right)^2 + \left(y_g^n - y_u^m\right)^2 + h_u^{m2}}$ is the Euclidean distance between $\mathrm{UAV}_m$ and $\mathrm{GU}_n$, $\eta_{\mathrm{LoS}}$ and $\eta_{\mathrm{NLoS}}$ represent the additional average path loss for LoS and NLoS links, respectively.

For A2G channel, let $\mathbf{P}_{\mathrm{LoS}}$ and $\mathbf{P}_{\mathrm{NLoS}}$ denote the probability of LoS and NLoS channel, $\mathbf{P}_{\mathrm{LoS}}$ is a simple modified sigmoid function, expressed as

$$\mathbf{P}_{\mathrm{LoS}} = \frac{1}{1 + a\exp\left(-b\left[\frac{180}{\pi}\theta - a\right]\right)} \tag{3}$$

where $a$ and $b$ are constants related to the given wireless propagation environment, $\theta$ is the elevation angle, which can be calculated by $\theta = \arcsin\left(h/d\left(\varpi_u^m, \varpi_g^n\right)\right)$. Consequently, the probability of occurrence for NLoS is given by $\mathbf{P}_{\mathrm{NLoS}} = 1 - \mathbf{P}_{\mathrm{LoS}}$. In this paper, we only deal with average path loss as ABS deployments are typically concerned with long-term variations of channel, not small scale fluctuations, small scale fading should be considered in instantaneous transmission links [52]. So the average path loss between $\mathrm{ABS}_m$ and $\mathrm{GU}_n$ can be given by

$$\overline{\mathcal{L}}_{m,n} = \mathbf{P}_{\mathrm{LoS}} \underbrace{d\left(\varpi_u^m, \varpi_g^n\right)\eta_{\mathrm{Los}}}_{\text{Los link path loss}} + \mathbf{P}_{\mathrm{NLoS}} \underbrace{d\left(\varpi_u^m, \varpi_g^n\right)\eta_{\mathrm{NLos}}}_{\text{NLos link path loss}} \tag{4}$$

For G2G channel, the path loss can be represented through the standard power loss model $\mathcal{L}_{v,n} = \mathrm{T}^{-1} d_{v,n}{}^{-\beta}$, where $T$ and $\beta$ are the additional path-loss and path-loss exponents, $v$ represents the serving GBS, $d$ is the propagation distance between GBS and GU.

Assuming that the ABS service range is a circular area with a radius of $R_{th}$ on the horizontal plane. We define the service range threshold of ABS through $\overline{\mathcal{L}}_{m,n}$. That is to say, when the distance between $\mathrm{ABS}_m$ and $\mathrm{GU}_n$ falls within a threshold $R_{th}$, the GU can be served by $\mathrm{ABS}_m$. Correspondingly, the problem is to find the optimal ABS altitude that maximizes $R_{th}$. The maximum path loss within the service area can be expressed as

$$\begin{aligned}\mathcal{L}_{max} = &\frac{A}{1 + a\exp\left(-b\left[\arctan\left(\frac{h}{R_{th}}\right) - a\right]\right)} \\ &+ 10\log\left(h^2 + R_{th}^2\right) + B\end{aligned} \tag{5}$$

where $A = \eta_{\mathrm{LoS}} - \eta_{\mathrm{NLoS}}$, and $B = 20\log f_c + 20\log\left(\frac{4\pi}{c}\right) + \eta_{\mathrm{NLoS}}$.

In order to obtain the optimal initial ABS altitude $h_{opt}$ and obtain the maximum coverage, we need to calculate $\frac{\partial R}{\partial h} = 0$ according to Eq. (5). By numerically solving this equation, we obtain the point where the radius–altitude curve changes its direction, that is the initial value of $h_{opt}$.

To enhance network utility, ABSs are equipped with mmWave antenna array to enhance signal strength in specific directions and isolate interference. Assuming $\mathfrak{D}_{\mathrm{a}}$ is the gain of ABS mmWave antenna. According to [53], it can be expressed as

$$\mathfrak{D}_{\mathrm{a}} = \frac{4\pi}{\Omega_A(\omega)}\cos(\varphi) \tag{6}$$

where $4\pi/\Omega_A(\omega)$ represents the maximum directivity of ABS antenna array, $\Omega_{\mathrm{a}}(\omega) = \pi/2\omega$ is the beam solid angle of the antenna array, $\omega$ is the directivity parameter of the antenna array, $\cos(\varphi)$ represents the radiation efficiency of the antenna array, and $\varphi \in [0, \pi/2]$ is the incident angle.

The antenna gain $\mathfrak{D}_{\mathrm{a}}$ depends on the directivity parameter $\omega$ and the radiation angle, and is symmetric along the vertical direction. It can be observed that the larger the $\omega$, the higher the directivity of antenna array, enabling it to concentrate on radiated power within a smaller area.

### 3.2. Performance metrics

Considering the realistic characteristics of HetECN, the main factors that constrain the effectiveness of it including the deployment delay of each ABS, along with network coverage rate. Deployment delay is a reflection of the response ability of ECN, and the coverage rate reflects the utility of HetECN.

In interference limited networks, the conventional method of defining coverage is through the signal-to-interference ratio (SIR), but due to the actual situation of HetECNs, unauthorized civilian signals and malicious jamming in military operations are inevitable. So we use signal-to-interference-plus-jamming ratio (SIJR) to measure network coverage performance. For G2G channel, the SIJR at $\mathrm{GU}_n$ (represented as $\mathcal{S}_{v,n}$) under jamming conditions can be expressed as

$$\mathcal{S}_{v,n} = \frac{p_G \cdot \mathcal{L}_{v,n}}{I_n + J_n}, v \in \mathcal{V}, n \in \mathcal{N} \tag{7}$$

For A2G channel, assuming a flat fading channel and the transmit power spectral density (PSD) of ABS is also flat. The $\mathrm{GU}_n$ is considered to be covered by $\mathrm{ABS}_m$ if SIJR (represented as $\mathcal{S}_{m,n}$) exceeds a preset threshold $\mathcal{S}_{th}$, and the SIJR received by $\mathrm{GU}_n$ can be given by

$$\mathcal{S}_{m,n} = \frac{p_A \cdot \overline{\mathcal{L}}_{m,n} \cdot \mathfrak{D}_{\mathrm{a}}}{I_n + J_n}, m \in \mathcal{M}, n \in \mathcal{N} \tag{8}$$

where $p_G$ and $p_A$ are the transmit power of GBS and ABS, $I_n = \sum_{i=1, i\neq m}^{M} p_A \cdot \overline{\mathcal{L}}_{m,n} + \sum_{j=1, j\neq n}^{N} p_G \cdot \mathcal{L}_{n,n'}$ is the sum of co-interference received by $\mathrm{GU}_n$ from the network, and $J_{\mathrm{n}}$ represents the jamming from AMJ. Since UAV systems are usually interference limited, we ignore the Gaussian noise.

Assuming the ABS employs frequency division multiple access (FDMA) technology, the total bandwidth occupied by $\mathrm{ABS}_m$ is $B_m = \sum_{n\in K} B_n$, where $K$ is the number of GUs served by $\mathrm{ABS}_m$. Therefore, the achievable rate for $\mathrm{GU}_n$ associated with $\mathrm{ABS}_m$ can be represented as

$$R_{m,n} = B_n \log_2\left(1 + \mathcal{S}_{m,n}\right), n \in K \tag{9}$$

Therefore, the total achievable transmission rate for HetECN can be expressed as $R_{total} = \sum_{n\in\mathcal{N}} B_n \log_2\left(1 + \mathcal{S}_{m,n}\right)$. Clearly, the sum of network rate of ABSs must exceed $R_{total}$, so we measure the required number of ABSs based on the minimum transmission rate needed by HetECN. thus, the required number of ABSs can be set as $M_{init} = \frac{R_{req}}{R_{SingleABS}}$. MmWave bandwidth is typically various from 30 to 300 GHz, for this paper, it is set at 38 GHz [54].

Denoting $\mathcal{T}_m \triangleq (0, \mathrm{T}_m)$ as the flight time of $\mathrm{ABS}_m$, let $\varpi_u'^m = \left\{x_u'^m(t), y_u'^m(t)\right\} t \in T$ represent the horizontal projection positions of $\mathrm{ABS}_m$, the trajectory length of $\mathrm{ABS}_m$ can be expressed as

$$L_m = \int_0^T \sqrt{1 + \left(\frac{y_u'^m(t)}{x_u'^m(t)}\right)^2} dx_u^m(t) \tag{10}$$

where $x_u'^m(t))$ and $y_u'^m(t))$ are the first derivatives of $x_u^m(t)$ and $y_u^m(t)$, respectively.

Typically, the trajectory of each ABS changes over time. For simplicity, the time domain $T_m$ is discretized into $t$ time slots, with each slot having a length of $\Delta t$, i.e. $T_m = t \cdot \Delta t$. Assuming the position of ABS updates every $\Delta t$. Thus, the ABS flight time model can be defined as

$$T_m^{total} = \sum_{t=0}^{T_m} \Delta t, \quad m \in \mathcal{M} \tag{11}$$

After reaching the optimal hovering position, ABS performs GU-ABS association based on the different traffic demands of GUs, ensuring the load balance of each ABS. Since each ABS has different flight trajectories, when measuring the performance of HetECN, we focus on the longest flight period of them.

Assuming the starting and ending positions of $\mathrm{ABS}_m$ are $s_m^{start} = \left(x_m^{start}, y_m^{start}, h_m^{start}\right)$ and $s_m^{end} = \left(x_m^{end}, y_m^{end}, h_m^{end}\right)$, respectively. When $\Delta t$ is sufficiently small, the position $\mathrm{ABS}_m$ within each slot is fixed. Under this assumption, the flight trajectory of $\mathrm{ABS}_m$ can be modeled as

$$\mathcal{E}_m = \left\{s_m^{start}, \ldots, s_m\left(t_i\right), \ldots, s_m^{end}\right\} \tag{12}$$

where $t_i \in T$, $m \in \mathcal{M}$.

To measure the performance of our proposed algorithm more intuitive, four additional evaluation metrics $\mathbb{U}, \mathbb{R}, \mathbb{L}$ and $\mathbb{A}$ are used in p performance evaluation. Specifically, $\mathbb{U}$ is the percentage of GUs without services, $\mathbb{R}$ is the average value of traffic load capacity. The

other two indicators are used to measure network balance, $\mathbb{L}$ is the coefficient of variation of traffic load, indicating the balance of traffic load capacity between ABSs, and $\mathbb{A}$ is the coefficient of variation of area, indicating the balance between service areas. The coefficient of variation $X$ is defined as the ratio of the system mean squared error (MSE) to the mean value, which can be calculated as

$$X = \frac{MSE}{\bar{X}}, X \in \{\mathbb{L}, \mathbb{A}\} \tag{13}$$

where $\bar{X}$ represents the mean value, and the value of coefficient of variation is inversely proportional to the degree of load balancing.

### 3.3. Problem formulation

In this work, we focus on achieving optimal utility of HetECN by optimizing ABS jamming avoidance trajectory and load balancing GU-ABS association strategy for heterogeneous GUs. In addition, logarithmic functions have the property of diminishing returns, which naturally achieves fairness between GUs, and we notice that utilizing the traffic fairness of each ABS can measure the degree of load balancing in HetECN. To this end, multi objective optimization problem $P1$ is established, where the network utility function is modeled as a logarithmic one, providing fair resource allocation among GUs. Therefore, The total load balancing utility of HetECN can be expressed as

$$\mathcal{U}\left(R_{total}\right) = \sum_{n\in N} \log\left(\sum_{m\in M} \mathcal{A}_{m,n} R_{m,n}\right) \tag{14}$$

$$\mathcal{A}_{m,n} = \begin{cases} 1 & n \in \theta_m \\ 0 & \text{otherwise} \end{cases}, \quad m \in \mathcal{M}, n \in \mathcal{N} \tag{15}$$

where $R_{m,n}$ is the achievable transmission rate of $\text{GU}_n$, $\mathcal{A}_{m,n}$ represents whether $\text{GU}_n$ is served by $\text{ABS}_m$, if $\text{GU}_n$ is associated with $\text{ABS}_m$, then $\mathcal{A}_{m,n} = 1$, otherwise, $\mathcal{A}_{m,n} = 0$. $\theta_m$ is the coalition of GUs served by $\text{ABS}_m$.

The optimization problem can be modeled as follows

$$\begin{aligned}
\mathcal{P}1: \ & \text{Given: } J, f(x,y), \mathcal{R}, \mathcal{M}, \mathcal{N}, v_{max}, F_{th}, \\
& \text{Find: } \mathcal{G}^*_{hov}, \Theta^{final} \\
& \text{Maximize: } \mathcal{U}\left(R_{total}\right) \\
& \text{Subject to: } R_m \geq \sum_{m\in M} \mathcal{A}_{m,n} \cdot R^{min}_{m,n}, \forall n \in \mathcal{N} && \text{(C1)} \\
& \mathcal{A}_{m,n} \in \{0,1\}, \forall m \in \mathcal{M}, n \in \mathcal{N} && \text{(C2)} \\
& \sum_{\text{m}=1}^{\text{M}} \mathcal{A}_{m,n} \leq 1, \forall m \in \mathcal{M}, n \in \mathcal{N} && \text{(C3)} \\
& \sum_{m=1}^{M} \left|N_m\right| = N, \forall m \in \mathcal{M} && \text{(C4)} \\
& h^{min}_u \leqslant h^m_u \leqslant h^{max}_u, \forall m \in \mathcal{M} && \text{(C5)} \\
& d_{m,n} \cdot \mathcal{A}_{m,n} \leq d_{max}, \forall m \in \mathcal{M}, n \in \mathcal{N} && \text{(C6)} \\
& T_{total} < T_{max} && \text{(C7)} \\
& 0 \leqslant N_m \leqslant 40, \forall m \in \mathcal{M} && \text{(C8)}
\end{aligned}$$

where $R^{min}_{m,n}$ represent the minimum rate demand of $\text{GU}_n$ associated with $\text{UAV}_m$, as shown in $C1$. $C2$ represents that the elements in the GU-ABS matching matrix are 0 or 1, $C3$ represents that a GU can only be associated with one ABS at most, $C4$ represents that the total number of GUs served by ABSs is no greater than the number of GUs in the network, $C5$ represents the ABS altitude constraint, $C6$ indicates that the distance from $\text{GU}_n$ to $\text{ABS}_m$ is less than $d_{max}$, and the variable $d_{m,n}$ represents the Euclidean distance between $\text{GU}_n$ and $\text{ABS}_m$. $C7$ represents the delay constraints of ABSs, $C8$ explains that each ABS can serve at most 40 GUs.

## 4. JA&LB algorithm

To effectively solving the optimal trajectory design and GU-ABS association problem, we propose a jamming avoidance and load balancing (JA&LB) algorithm, which decompose problem $P1$ into three sub problems, i.e., ABS trajectory design, load balancing, and power consumption optimization. Firstly, solving the problem of ABS jamming avoidance trajectory design, and then optimize the GU-ABS association based on load balancing according to various GU demands. Finally, power consumption optimization is also carried out using ABS altitude fine-tuning. The overall flowchart of JA&LB algorithm is shown in Fig. 2.

### 4.1. JAA algorithm for trajectory design

In this section, an improved jamming avoidance ABS trajectory planning method based on improved APF algorithm (JAA) is developed to solve the proposed problem. Considering the initial position $\varpi^{start}_u = \left\{x^{start}_u, y^{start}_u, h^{start}_u\right\}$ of ABSs and different demand constraints of GUs, we design a ABS trajectory planning scheme relevant to real-world scenarios in HetECN with multiple obstacles, active jamming and other constraints, which can efficiently finding the optimal ABS trajectory in HetECN.

#### 4.1.1. Introduction of APF method

The basic APF algorithm defines the attractive force field function and repulsive force field function. Say concretely, the attractive force field function can be represented as

$$U_{att,standard}(X) = \frac{1}{2} K_{att}\left(X - X_G\right) \tag{16}$$

where $K_{att} > 0$ represents the coefficient constant of attraction field, $X_G$ denotes the position of target. The attraction force can be expressed as the gradient of attraction field function, which can be derived as

$$F_{att}(X) = -\nabla\left(U_{att,standard}(X)\right) \tag{17}$$

The repulsive force field function can be expressed as

$$U_{rep,standard}(X) = \begin{cases} \frac{1}{2} K_{rep}\left(\frac{1}{\rho(X,X_{obs})} - \frac{1}{\rho_0}\right)^2, \\ \qquad \rho\left(X, X_{obs}\right) \leq \rho_0 \\ 0, \quad \rho\left(X, X_{obs}\right) > \rho_0 \end{cases} \tag{18}$$

where $K_{rep} > 0$ represents the coefficient constant of the repulsive field, $\rho_0$ denotes the maximum range of the obstacle's repulsion field, and $\rho\left(X, X_{obs}\right)$ is the closest distance between specific object and obstacle.

Therefore, the repulsive force can be expressed as the negative gradient of repulsive force field function as

$$F_{rep}(X) = -\nabla\left(U_{rep,standard}(X)\right) \tag{19}$$

Therefore, the total force experienced by specific object is

$$F(X) = F_{rep}(X) + F_{att}(X) \tag{20}$$

When applied to ABS trajectory design, the force between GU and ABS is modeled as attraction force, while the force between obstacles, interference and ABS are mutual repulsion force, respectively.

#### 4.1.2. ABS trajectory design based on improved APF

Due to the complexity environment in HetECN, both civilian devices in disaster rescue and active electromagnetic jamming in military operations are inevitable, ABSs should also ensure that they do not fly out the mission area. So based on the traditional APF method, four additional virtual forces are defined, which are introduced as follows.

**Jamming repulsion force (JRF)**. When the ABS is within the range of active jamming in HetECN, it will fly away from the direction of AMJ

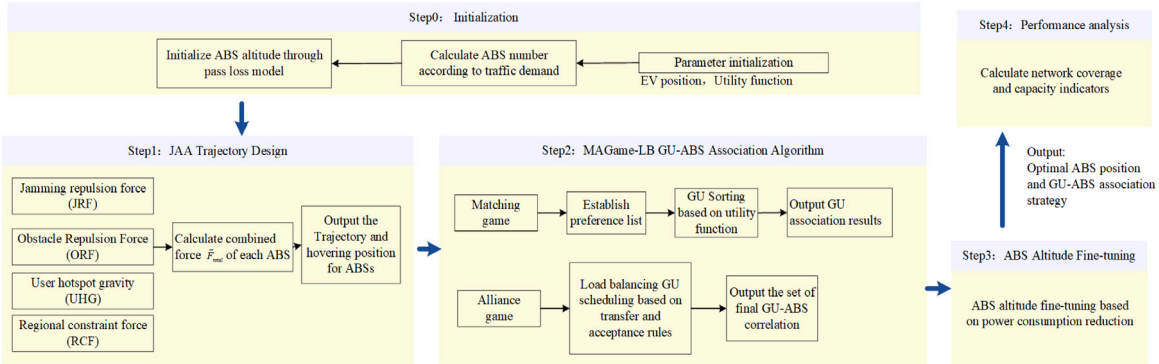



**Fig. 2.** Overall flowchart of proposed JA&LB algorithm.

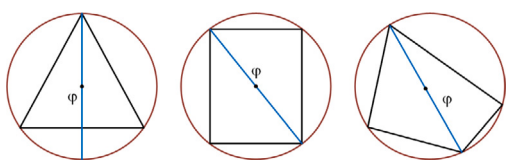



**Fig. 3.** Simplified obstacle model.

to decrease the influence of it, the magnitude of $\overrightarrow{JRF}$ can be expressed as

$$\left|\overrightarrow{JRF}\right|_{rep} = -\nabla\left(U_{rep}(X)\right) = \begin{cases} K_{rep}\left(\frac{1}{\rho(X,X_j)} - \frac{1}{\rho_j}\right)\frac{1}{\rho^2(X,X_j)}\frac{\partial\rho(X,X_j)}{\partial X} & ,\rho\left(X,X_j\right) \leqslant \rho_j \\ 0 & ,\rho\left(X,X_j\right) > \rho_j \end{cases} \tag{21}$$

where $X_j$ and $\rho_j$ represent the coordinate and maximum influence distance of AMJ, $K_{rep}$ is the repulsive potential field coefficient, $\rho\left(X,X_j\right) = \left\|X - X_j\right\|$ indicates the Euclidean distance between ABS and AMJ.

**Obstacle repulsion force (ORF).** In actual flight of ABSs, the obstacles encountered are mostly irregular 3D objects, if the repulsive potential field is calculated along the boundary points of obstacles at each waypoint, it will lead to frequent turns of ABSs. To address this issue, we vertically projects the contours of all obstacles, with the circumscribed circles representing them. Simplify the complex real environment to a 2D model. With the diameter $\varphi$ of the circumscribed circles representing the length of connecting segment between two farthest points on the object contour, as shown in Fig. 3.

Introducing the relative distance between ABS and target into the repulsion field function, and the improved repulsion field function is established as follows

$$U_{rep}(X) = \begin{cases} \frac{1}{2}K_{rep}\left(\frac{1}{\rho(X,X_o)} - \frac{1}{\rho_0}\right)^2 \cdot \rho^{\alpha}\left(X,X_o\right), \\ \qquad \rho\left(X,X_o\right) \leq \rho_0 \\ 0, \quad \rho\left(X,X_o\right) > \rho_0 \end{cases} \tag{22}$$

where $\alpha$ is the regulatory factor, which is a variable normal number, $\rho_0$ is the range of the repulsive field of the obstacle, beyond which the repulsive field no longer acts. Taking the negative gradient of the repulsion field, the repulsion function can be obtained as

$$|\overrightarrow{ORF}|_{rep} = \begin{cases} |\overrightarrow{ORF}|_{rep1} + |\overrightarrow{ORF}|_{rep2} & ,\rho\left(X,X_o\right) \leqslant \rho_o \\ 0 & ,\rho\left(X,X_o\right) > \rho_o \end{cases} \tag{23}$$

where the specific forms of $|\overrightarrow{ORF}|_{rep1}$ and $|\overrightarrow{ORF}|_{rep2}$ are

$$|\overrightarrow{ORF}|_{rep1} = K_{rep}\left(\frac{1}{\rho\left(X,X_o\right)} - \frac{1}{\rho_0}\right)\rho^{\alpha-2}\left(X,X_o\right) \tag{24}$$

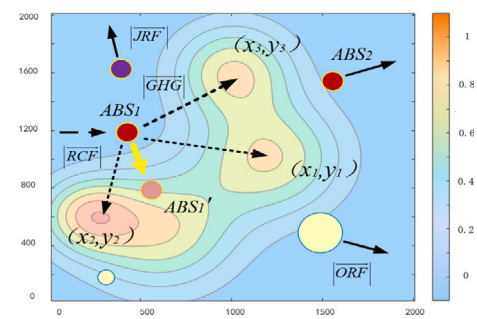



**Fig. 4.** Schematic diagram of forces in APF.

$$|\overrightarrow{ORF}|_{rep2} = -\frac{1}{2}\alpha K_{rep}\left(\frac{1}{\rho\left(X,X_o\right)} - \frac{1}{\rho_0}\right)^2\rho^{\alpha-1}\left(X,X_o\right) \tag{25}$$

where $|\overrightarrow{ORF}|_{rep1}$ and $|\overrightarrow{ORF}|_{rep2}$ are the two reverse direction components of the repulsive force, the regulatory factor $\alpha \in (0,1)$ alters the magnitude of force as the distance between ABS and obstacle changes.

**GU hotspot gravity (GHG).** According to the factor that GU hotspots result in a stronger attraction force at that point, we also modeled the gravity of GU hotspots in HetECN. Yet due to the fact that any point within the APF will generate an attractive force for ABS, it is difficult to calculate the total combined force of countless GUs. Therefore, we uniformly sample points of GUs to measure the resultant force. We define $\overrightarrow{GHG}$ as the hotspot gravity generated by GU hotspot area, its magnitude can be expressed as

$$\left|\overrightarrow{GHG}\right|_{att} = K_{att}\sqrt{\left(x - x_s\right)^2 + \left(y - y_s\right)^2} \tag{26}$$

where $K_{att}$ is the attraction potential gain coefficient, $\left(x_s, y_s\right)$ represents the coordinates of the sampling point.

Therefore, the combined force generated by GU can be described as

$$\overrightarrow{F}_{att} = \sum_{j=1}^{P}\left(\left|\overrightarrow{GHG}\right|\right) \tag{27}$$

where $P$ is the set of sampling points within the maximum influence distance of typical ABS in the network.

**Regional constraint force (RCF).** Typically in rescue or military operations, tasks are executed within predefined task areas. Therefore, APF should include boundary constraints to ensure ABS do not flight out of task area. The $\overrightarrow{RCF}$ is also calculated through gradient as

$$\left|\overrightarrow{RCF}\right|_{rep} = -\nabla\left(U_{rep}(X)\right) = \begin{cases} K_{rep}\left(\frac{1}{\rho(X,X_b)} - \frac{1}{\rho_p}\right)\frac{1}{\rho^2(X,X_b)}\frac{\partial\rho(X,X_b)}{\partial X} & ,\rho\left(X,X_b\right) \leqslant \rho_b \\ 0 & ,\rho\left(X,X_b\right) > \rho_b \end{cases} \tag{28}$$

where $\rho_b$ represents the maximum influence distance of the mission area boundary, $\rho\left(X, X_b\right)$ is the closest distance from ABS to the boundary of task area.

Fig. 4 shows the combined force in HetECN, $\mathrm{ABS}_1'$ will be the next waypoint for $\mathrm{ABS}_1$. Therefore, the direction of the resultant force acting on ABS at each waypoint determines the direction of ABS's movement in the next time slot, and the combined force ABS suffered can be expressed as

$$\begin{aligned}\overrightarrow{F}_{total} &= \overrightarrow{F}_{att} + \overrightarrow{F}_{rep} \\ &= \sum_{j=1}^{P}(|\overline{GHG}|) + \left(\overline{ORF} + \overline{JRF} + \overline{RCF}\right)\end{aligned} \tag{29}$$

The main idea of JAA is to find a position that can maintain force balance, i.e., zero resultant force, and that position is the optimal hovering position of ABS. However, it is difficult to guarantee that the total resultant force achieve zero. Therefore, we assume that when the total resultant force is less than the given threshold $F_{th}$, the stop condition is achieved.

Since the combined force can change the flight speed and direction of ABSs, it is necessary to establish a mapping between $\vec{F}$ and velocity $v$. We use the trigonometric function $\arctan(\cdot)$ to describe this relationship as

$$\vec{v} = \frac{2v_{max}}{\pi} \cdot \arctan(\vec{F}) \tag{30}$$

where $\arctan(\vec{F})$ changes within $[0, \pi/2]$, corresponds to the velocity change at $\left[0, v_{max}\right]$.

Then, the ABS will move to the next waypoint within time slot $\Delta t$, and then update the resultant force to support ABS in finding the next one. In an ideal scenario, when the total resultant force is zero, ABS will stop position updating. Following the resultant force, each ABS updates its position until the entire network is in a stable topology.

The pseudo code of JAA algorithm is summarized in Algorithm 1.

**Algorithm 1:** Jamming avoidance UAV path planning method based on APF, JAA

**Input:** $J, f(x, y), \mathcal{R}, \mathcal{M}, \mathcal{N}, v_{max}, F_{th}$
**Output:** $\mathcal{G}_{path} = \left[\mathcal{G}_{path}^1, \mathcal{G}_{path}^2, \cdots, \mathcal{G}_{path}^M\right]$,
$\mathcal{G}_{hov} = \left[\varpi_{hov}^1, \varpi_{hov}^2, \cdots, \varpi_{hov}^M\right]$, $m \in \mathcal{M}$,

```
initialization;
while |F_total| > F_th do
    Calculate the combined force F_total of ABSs based on
      equation Eq.(29) at current time slot;
    Update the step size v for each ABS according to equation
      Eq.(30);
    Calculate the next waypoint of each ABS;
    if |F_total| ⩽ F_th or |F_total| = 0 then
        stop JAA algorithm;
        Output ABS trajectory and hover position set;
    else
        t=t+1;
        repeat;
    end
end
```

In each iteration of Algorithm 1, the ABS determines the direction of movement for the next time step $\Delta t$ based on the resultant force, and finally outputs the set of ABS waypoints $\mathcal{G}_{path}$, as well as the final hovering position set $\mathcal{G}_{hov}$ of all ABSs.

### 4.2. MAGame-LB algorithm for GU-ABS association

In this section, we apply game theory to solve the load balancing GU association problem in HetECN. Specifically, we propose a matching and alliance game based on load balancing (MAGame-LB) algorithm, which transforms the problem into a many to one matching sub game and an alliance sub game, ultimately map them to the proposed GU-ABS association problem.

#### 4.2.1. Matching game and related concepts

We transform the GU-ABS association problem in $P1$ into a bilateral matching game, the sets of ABSs and GUs are both rational individual agents, each choosing their own decision and attempting to maximize their individual utility in the distributed matching process. ABSs and GUs are divided into clusters, where one ABS allows simultaneous service of multiple GUs within the same cluster, but each GU can only enter one cluster at most, the matching game theory can effectively solve this problem [42].

**Definition 1** (*Matching*)**.** Given the sum of two disjoint sets of game players $\mathcal{M}$ and $\mathcal{N}$, a matching $\Phi$ is a bidirectional mapping $\mathcal{M} \leftrightarrow \mathcal{N}$, such that for all $m \in \mathcal{M}$ and $\mathrm{n} \in \mathcal{N}$, there is

(a) $\Gamma(n) = m \Leftrightarrow \Gamma(m) = n$,

(b) $|\Gamma(n)| \leq 1, |\Gamma(\mathrm{m})| \leq N$.

where $|\Gamma(\cdot)|$ represents the number of elements in the matching result $\Gamma(\cdot)$, (a) indicates that ABS and GU matches each other, (b) indicates that each GU can match up to one ABS, while each ABS can serve multiple GUs. When $\mathrm{GU}_n$ are not allowed to use $\mathrm{ABS}_m$ or $\mathrm{ABS}_m$ is not assigned to $\mathrm{GU}_n$, we have $\Gamma(\mathrm{m}) = \varnothing$ or $\Gamma(\mathrm{n}) = \varnothing$.

In MGame-LB algorithm, utility function is used to measure the degree of matching between the two players. Assuming that the utility when $\mathrm{GU}_n$ is connected to $\mathrm{ABS}_m$ and the utility when $\mathrm{ABS}_m$ is connected to $\mathrm{GU}_n$ are defined as $\tilde{\mathcal{U}}_n^m$ and $\tilde{\mathcal{U}}_m^n$, respectively. But in practice, before all GUs and ABSs make their selections, the exact interference that an GU suffers is unavailable and inevitable, which is known as externality. So the estimated utility function is used when $\mathrm{GU}_n$ is connected to $\mathrm{ABS}_m$ can be expressed as

$$\tilde{\mathcal{U}}_n^m = \log\left[B\log_2\left(1 + \tilde{S}_{m,n}\right)\right] \tag{31}$$

where $\tilde{S}_{m,n} = \frac{p_n^A \cdot \overline{\mathcal{L}}_{m,n} \cdot \mathfrak{D}_{\mathrm{a}}}{\sum_{i=1, i\neq m}^{M} p_A \cdot \overline{\mathcal{L}}_{m,n} + J_n}$ only depends on the interference from the ABS and AMJ.

Due to the various traffic demands of GUs, the influence of each GU on the total utility of HetECN is different. Therefore, ABS will accept limited number of GUs that have the maximum increment for their total utility. The incremental utility function when $\mathrm{ABS}_m$ is connected to $\mathrm{GU}_n$ is expressed as

$$\tilde{\mathcal{U}}_m^n = \sum_{n\in\mathcal{N}_{\mathrm{m}}'} \tilde{\mathcal{U}}_n^m - \sum_{n\in\mathcal{N}_{\mathrm{m}}'\{n'\}} \tilde{\mathcal{U}}_n^m \tag{32}$$

where $n'$ is the newly added GU to the cluster, $\mathcal{N}_{\mathrm{m}}'$ is the set of GUs after adding $n'$, and $\tilde{\mathcal{U}}_m^n$ is the incremental utility contributed by $\mathrm{GU}_n$ to $\mathrm{ABS}_m$. There is a gap between the outcome of matching based on estimated utilities and that of matching based on accurate utilities. However, the gap will be narrowed through AGame-LB sub game in Section 4.2.3.

In the process of sorting ABS or GU, preference relationship is used to construct a preference list between them.

**Definition 2** (*Preference List*)**.** The preference relationship $\succ_{\mathrm{i}}$, $i \in \mathcal{M} \cup \mathcal{N}$ is used to reflect the full order of preferences of GU, that is to say, for any two GUs, we have

$$n_1 \succ_m n_2 \Leftrightarrow \mathcal{U}_m^{n_1} > \mathcal{U}_m^{n_2}, \quad n_1, n_2 \in \mathcal{N} \tag{33}$$

where $m_1, m_2 \in \mathcal{M}$, $n_1, n_2 \in \mathcal{N}$, $n_1 \neq n_2$.

Similarly, defining a preference relationship $\succ_m$ based on the utility of two GUs in the set of $\mathrm{ABS}_m$, such that for any two ABSs, we can obtain

$$m_1 \succ_n m_2 \Leftrightarrow \mathcal{U}_n^{m_1} > \mathcal{U}_n^{m_2}, \quad m_1, m_2 \in \mathcal{M} \tag{34}$$

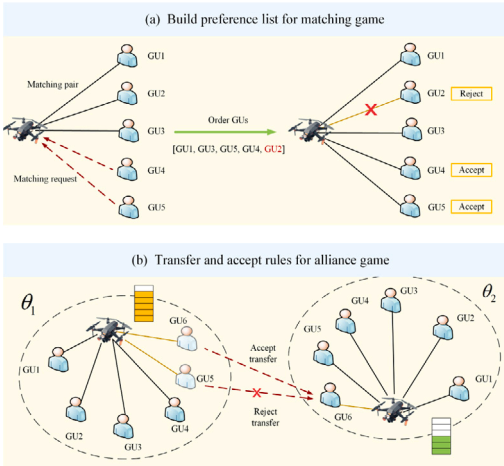



**Fig. 5.** Stepwise synthesis of two game theory methods.

As shown in Fig. 5(a), by broadcasting preference relationships and necessary information through EV, each GU will sort all ABSs in non increasing order based on the preference relationship in Eq. (33), then establish their preference list. Due to the limited capacity of ABS, it cannot accept all applications for GUs. Therefore, it is also necessary for ABS to establish preference list to select GUs. Only the top ranked GU in the preference list will be accepted, while other applicants will be rejected and proceed to the next round, rebuilding their preference list.

**Definition 3** (*Matching Game*)**.** A matching game is defined by a triplet $(\mathcal{M}\times\mathcal{N}, \succ_{\mathcal{M}}, \succ_{\mathcal{N}})$, where $\succ_{\mathcal{M}} = \left\{\succ_{m_i}\right\}_{m_i\in\mathcal{M}}$ and $\succ_{\mathcal{N}} = \left\{\succ_{n_j}\right\}_{n_j\in\mathcal{N}}$ represents the preference relationship ranking of ABSs and GUs, respectively. Matching game is a reflexive, complete, and transitive binary relationship established by participants. A typical process of it is in Fig. 5.

It is not difficult to see that the solution of a matching game is defined as a match $\Phi$ on the set $\mathcal{M}\times\mathcal{N}$, and then we propose to find a stable solution for the matching game.

**Definition 4** (*Stable Matching*)**.** If there exit no simultaneous allocation to any pair of $\mathrm{GU}_n^1, \mathrm{GU}_n^2 \in \mathcal{N}$ allocated to $\mathrm{ABS}_m^1, \mathrm{ABS}_m^2 \in \mathcal{M}$ that satisfies the following conditions

(a) $\mathrm{GU}_n^2$ prefer $\mathrm{ABS}_m^1$ rather than $\mathrm{ABS}_m^2$, i.e., $\mathrm{ABS}_m^1 \succ_{n_2} ABS_m^2$, $\mathrm{ABS}_m^1$ prefer $\mathrm{GU}_n^2$ rather than $\mathrm{GU}_n^1$, i.e., $\mathrm{GU}_n^2 \succ_{m_1} \mathrm{GU}_n^1$,

(b) If $\mathrm{GU}_n^2$ replaced $\mathrm{GU}_n^1$, all GUs in $\mathrm{ABS}_m^1$ meet all constraints in $P1$.

The match $\Phi$ is considered to be a stable matching $\Phi^*$.

#### 4.2.2. MGame-LB algorithm for GU-ABS matching

Due to the externality of HetECN, we propose a GU-ABS matching game based on load balancing (MGame-LB) to find the stable solution of this sub game. The pseudo code for MGame-LB is shown in Algorithm 2.

Specifically, the MGame-LB algorithm constructs its preference prototype through the preference relationship between ABS and GU according to Eqs. (33) and (34). When some GUs send applications to $\mathrm{ABS}_m$, $\mathrm{ABS}_m$ will sort the applicants based on the estimated preference utility function in Eq. (31). It is worth noting that when multiple GUs within the distance constraint apply for the same $\mathrm{ABS}_m$ service, $\mathrm{ABS}_m$ will accept the most preferred one that satisfies the constraint in $P1$ and reject the others. Then, the GU be rejected reapplies for their next best choice ABS based on their preference list. Correspondingly, the relevant ABS updates its list. Finally obtain the optimal sets $\Phi^*$ of GU-ABS matching.

**Algorithm 2:** Matching game based on load balancing GU scheduling, MGame-LB

**Input:** $\mathcal{M}, \mathcal{N}$
**Output:** $\Phi^*$
initialization;
**for** $n \in \mathcal{N}$ **do**
  Establish preference list based on Eq.(31) and Eq.(33);
**end**
**while** $\mathcal{N} \neq \varnothing$ **do**
  Calculate the most preferred ABS;
  **for** $\mathrm{ABS}_m \in \mathcal{M}$ **do**
    Establish preference list based on Eq.(32) and Eq.(34);
    Sort application GUs based on preference list;
    Check from the most preferred applicants. If the constraint conditions in $P1$ are met, add the GU to the waiting list, otherwise it will be rejected. When multiple GUs meet the associated constraint conditions, ABS only retains the optimal GU;
  **end**
  Update user association matching results $\Phi^*$;
  Update $\mathcal{N}$ to reject $n \to \mathcal{N}$;
**end**

**Property 1.** *The GU-ABS matching $\Phi^*$ obtained from MGame-LB algorithm is a stable one.*

**Proof.** Firstly, the preference list of GU applicant is finite, so the MGame-LB algorithm is convergent. Secondly, each ABS can serve a finite number of GUs within its neighborhood in a constraint distance $d_{max}$, so the possible number of GU transfers is also limited. Suppose there is a situation where $\mathrm{GU}_n^1, \mathrm{GU}_n^2 \in \mathcal{N}$ are assigned to $\mathrm{ABS}_m^1$ and $\mathrm{ABS}_m^2$, respectively. If $\mathrm{GU}_n^2$ is accepted instead of $\mathrm{GU}_n^1$, we have $\mathrm{GU}_n^2$ preferring $\mathrm{ABS}_m^1$ instead of $\mathrm{ABS}_m^2$, $\mathrm{ABS}_m^1$ preferring $\mathrm{GU}_n^2$ instead of $\mathrm{GU}_n^1$, and constraint in $P1$ holds for all GUs in $\mathrm{ABS}_m^1$. Then, according to algorithm 2, assign $\mathrm{GU}_n^2$ to $\mathrm{ABS}_m^1$. However, this contradicts the fact that $\mathrm{GU}_n^2$ is assigned to $\mathrm{ABS}_m^2$. Therefore, the GU-ABS matching $\Phi^*$ generated by MGame-LB algorithm is stable.

**Property 2.** *The complexity of MGame-LB algorithm in the worst case is $\mathcal{O}(MN)^3$.*

**Proof.** Noticing that the first for loop in Algorithm 2 is related to the establishment of preference list of GU and the sorting of each ABS, with a complexity of $\mathcal{O}((M+N)\left(M+M^2\right))$. In addition, in each iteration of the while loop, the second for loop is related to the calculation of ABS's preference function, sorting and constraint checking of each GU, with a complexity of $\mathcal{O}(MN + MN^2 + 4MN) = \mathcal{O}(5MN + MN^2)$. Since the allocation of GU in all networks requires at most $N$ iterations, the computational complexity of the while loop should be $\mathcal{O}(N\left(5MN + MN^2\right))$. Therefore, in the worst-case scenario, the complexity of MGame-LB is $\mathcal{O}(MN)^3$. Therefore, using MGame-LB algorithm, the matching game can achieve stable matching $\Phi^*$ with lower computational complexity.

#### 4.2.3. Alliance game and related concepts

Due to the fact that the estimated utility function in Eq. (31) is used to construct the preference list for GU-ABS matching sub game, avoiding ubiquitous interference in HetECN, which will generate deviation with actual HetECN scenarios. To solve this issue caused by externalities, we propose a GU-ABS alliance game based on load balancing (AGame-LB). Specifically, due to the fact that the more GUs one ABS serves, the greater the various interference it received, we assume that GUs are selfish and rational, they cannot all be served

by the same ABS. On the contrary, GUs tends to choose a ABS with larger bandwidth and less interference. In this case, the process of GUs choosing ABS can be described as an alliance game.

**Definition 5** (*Alliance Game*)**.** The alliance game is characterized by element pairs $(\Theta, \Psi)$, $\Theta = \{\theta_1, \theta_2, \ldots, \theta_M\}$ is the alliance partition between ABS and GU, $\Psi$ is a mapping between each alliance $\theta_m \in \Theta$ and utility function $\mathcal{U}(\theta_m)$.

Generally speaking, the optimal division or solution of an alliance game is obtained through exhaustive search. However, the number of possible alliances increases exponentially with the increase number of GUs [43], making exhaustive search impractical. In this case, a suboptimal algorithm with low computational complexity is proposed instead.

#### 4.2.4. AGame-LB algorithm for GU-ABS alliance

To better understand the choices of GU and ABS, the transfer and acceptance rules are introduced as follows.

**Definition 6** (*Transfer Rules*)**.** Due to the fact that moving some GUs from heavy load ABS to light load one can improve the utility of network, given the alliance partition $\Theta$, if

(a) The utility $\mathcal{U}(\theta_j)$ of GU in alliance $\theta_j$ is greater than $\mathcal{U}(\theta_i)$ in alliance $\theta_i$, $j, i \in \mathcal{M}$,

(b) $\mathrm{GU}_n$ is been accepted by alliance $\theta_j$.

Then $\mathrm{GU}_n$ in alliance $\theta_i$ can be transferred to alliance $\theta_j$.

**Definition 7** (*Accept Rules*)**.** For any GU actually accepted by alliance $\theta_j$, the following conditions must be satisfied

$$\begin{aligned} &\mathcal{U}_i(\theta_i') + \mathcal{U}_j(\theta_j') + \sum_{m \in \mathcal{M}\{i,j\}} \mathcal{U}_m'(\theta_m) \\ &\quad > \mathcal{U}_m(\theta_m) + \mathcal{U}_j(\theta_j) + \sum_{m \in \mathcal{M}\{i,j\}} \mathcal{U}_m(\theta_m) \end{aligned} \tag{35}$$

where $\vartheta_i' = \theta_i \backslash \{n\}$, $\theta_j' = \theta_j \cup \{n\}$ and $\mathcal{U}_m(\theta_m)$ are the utility of $(\theta_m)$ after GU divert from $\theta_i$ to $\theta_j$. This condition ensures that the transfer increases total utility of network.

When multiple GUs simultaneously apply to transfer to the same ABS, it is necessary to sort the priorities of them. To characterize preference of ABS, a utility factor $\delta$ is defined to help participants in $\theta_m$ accept the optimal one. The definition of $\delta$ is as follows

$$\begin{aligned} \delta(\theta_m, \theta_j) = &\sum_{m \in \mathcal{M} \backslash \{i,j\}} \left(\mathcal{U}_m'(\theta_m) - \mathcal{U}_m(\theta_m)\right) \\ &\times \mathcal{U}_m(\theta_m') + \mathcal{U}_j(\theta_j') - \mathcal{U}_m(\theta_m) - \mathcal{U}_j(\theta_j) \end{aligned} \tag{36}$$

As shown in Fig. 5(b), joint using transfer rules, acceptance rules and utility factor, the AGame-LB algorithm is proposed to determine the final match of GU-ABS association. Specifically, at the beginning of AGame-LB, an initial partition is set $\Theta^{ini} = \{\theta_1^{ini}, \theta_2^{ini}, \ldots, \theta_M^{ini}\}$ according to algorithm 2. Then, each GU indicates their preference to transfer to another alliance. If the transfer satisfies the transfer rule, each alliance filters the transfer applicants based on the acceptance rule. The algorithm selects the applicant with the largest value of $\delta$ and rejects other GUs. Let $t$ be the number of iterations to form a new GU-ABS association set $\Theta^t = \{\theta_1^t, \theta_2^t, \ldots, \theta_M^t\}$, repeat this process until algorithm convergence, i.e. $\left\|\Theta^t - \Theta^{t-1}\right\|^2 < \varepsilon$, where $\varepsilon$ is a predetermined threshold, $\varepsilon_1$ is the threshold after iteration. Finally, we map the final partition $\Theta^{final}$ to the result of GU-ABS association problem defined in $P1$.

The pseudo code of AGame-LB is summarized in Algorithm 3.

**Property 3.** *The AGame-LB algorithm will eventually converge to a final set $\Theta^{final}$.*

**Algorithm 3:** GU-ABS Alliance Game Based on Load Balancing, AGame-LB

**Input:** t=0, $\Theta^0 = \Theta^{ini}, \varepsilon, \varepsilon_1$
**Output:** $\Theta^{final} = \Theta^t$

1 **while** $\varepsilon_1 > \varepsilon$ **do**
2  **for** $n \in \mathcal{N}$ **do**
3  According to the transfer rule defined in Definition 6, calculate GU's most preferred transfer;
4  **end**
5  **for** $\theta_m^t \in \Theta^t$ **do**
6  Screen GU applicants according to the acceptance rules defined in Definition 7;
7  Select the application GU with the highest transfer factor defined in Eq.(36) and reject the remaining application GUs;
8  Use $\theta_m^t$ and newly accept GU to update $\theta_m^{t+1}$;
9  **end**
10  Calculate $\varepsilon_1 = \left\|\Theta^t - \Theta^{t-1}\right\|^2$
11  Update $\Theta^t = \{\theta_1^t, \theta_2^t, \ldots, \theta_M^t\}$;
12  t=t+1;
13 **end**

**Proof.** Given an initial partition $\Theta^{ini}$, the alliance building actions can be regarded as a sequence of transfer operations, i.e.,

$$\Theta^{ini} \to \Theta^1 \to \Theta^2 \to \cdots \to \Theta^t \to \Theta^{t'} \cdots \tag{37}$$

where $\Theta^t = \{\theta_1^t, \theta_2^t, \ldots, \theta_M^t\}$ is the alliance set formed after $T$ iterations.

According to the definitions of transition and acceptance rules, when $\mathrm{GU}_n$ transitions from $\theta_i$ to $\theta_j$, we have $\mathcal{U}(\theta_i \backslash \{n\}) + \mathcal{U}(\theta_j \cup \{n\}) > \mathcal{U}(\theta_i) + \mathcal{U}(\theta_j)$. On the other hand, due to the absence of interference between different ABSs, the transfer of $\mathrm{GU}_n$ between alliances $\theta_i$ and $\theta_j$ does not affect the utility of other alliances in $\Theta^t \backslash \{\theta_i, \theta_j\}$. Therefore, for any transition $\Theta^t \to \Theta^{t'}$, the following inequality always holds

$$\sum_{\theta_i \in \Theta^{t'}} \mathcal{U}(\theta_i) > \sum_{\theta_i \in \Theta^t} \mathcal{U}(\theta_i) \tag{38}$$

The above equation indicates that the transfer of GU-ABS association increases the overall utility of network, and $\Theta^t \neq \Theta^{t'}$ always holds for any transfer $\Theta^t \to \Theta^{t'}$ $(\mathrm{t} \neq \mathrm{t}')$, thus ensuring that the transfer sequence in Eq. (37) converges to a final set $\Theta^{final}$.

**Property 4.** *The complexity of AGame-LB is $\mathcal{O}(TMN)$.*

**Proof.** In each iteration of the while loop in step 2, when GU obtains the most preferred association rules, the complexity of both in the first for loop is $\mathcal{O}(MN)$. For the GU applicant with the maximum transfer factor $\delta$, the complexity of the second for loop is $\mathcal{O}((7+M)N)$, and the computational complexity of the while loop is $\mathcal{O}(T[2MN+(7+M)N]) = \mathcal{O}(T(7+3M)N)$. Therefore, the complexity of the AGame-LB algorithm is $\mathcal{O}(TMN)$.

#### 4.2.5. Algorithm for load balancing GU-ABS association problem

By sequential using the above two game theory methods, the MAGame-LB algorithm proposed for jamming avoidance ABS trajectory design and load balancing GU-ABS association in problem $P1$ is shown in Algorithm 4.

### 4.3. ABS altitude fine-tuning method

Obviously, compared to other scenarios, HetECN often suffer higher path losses. As mentioned earlier, the maximum coverage radius $R$ of ABS first increases and then decreases with the increase of ABS

**Algorithm 4:** Matching and Alliance Game Based on Load Balanced GU Scheduling, MAGame-LB

**Input:** $\mathcal{M}, \mathcal{N}$, t = 0, $\Theta^0 = \Theta^{ini}, \varepsilon, \varepsilon_1$

1 Obtain a GU-ABS stable matching $\Phi^*$ through Algorithm 2;
2 Mapping $\Phi^*$ to $\Theta^{ini}$, i.e., $\Phi^* \to \Theta^{ini}$;
3 Using Algorithm 3 and the results of Step 2 to obtain the final division of GU-ABS alliance $\Theta^{final}$;
4 Mapping $\Theta^{final}$ to $\mathcal{M} \times \mathcal{N}$;
5 $\mathcal{M} \times \mathcal{N}$ is the solution to the load balancing and GU-ABS association problem defined in $P1$.

**Output:** $\Theta^{final}$

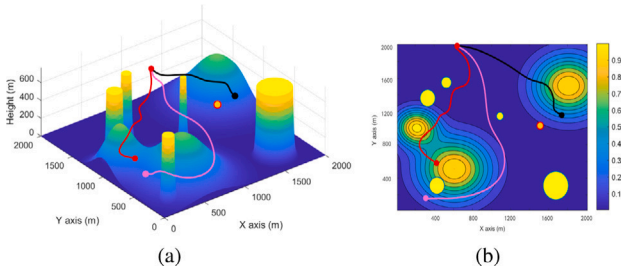



**Fig. 6.** ABS trajectory. (a) 3D perspective; (b) vertical view.

flight altitude. Therefore, we can adjust the altitude of each ABS to increase the probability of LoS transmission, along with reduce ABS power consumption without reducing network utility [5].

We preset the initial altitude of ABS as the inflection point of the curve, which means that ABS can provide the maximum coverage radius that meets the path loss requirements. At this point, if the service radius of ABS is shortened based on its actual service area, the altitude can be reduced while satisfying the path loss model, thereby reducing ABS power consumption. Therefore, after obtaining the final horizontal position of ABS as $\mathcal{G}_{hov} = \left[\varpi^1_{hov}, \varpi^2_{hov}, \ldots, \varpi^M_{hov}\right]$, if the distance between the farthest GU served by $\text{ABS}_m$ and the horizontal projection of it is less than the maximum coverage radius, i.e., $\max\left\{u_i(x), x \in R_i\right\} < R_{max}$, the ABS altitude can be reduced. For each $\text{ABS}_m$, let the actual coverage radius be $R_a = \max\left\{u_i(x), x \in R_i\right\}$ and the maximum path loss be $\rho$, The final altitude of ABS can be obtained as

$$\overline{\mathcal{L}}_m\left(h^*, R_a\right) = \rho \tag{39}$$

By solving the above equation, the optimal ABS hovering positions $\mathcal{G}^*_{hov}$ can be obtained that satisfies $P1$. And then the optimal problem in $P1$ is completely completed.

## 5. Simulation and results

In this section, simulations results are provided to validate the performance of the proposed $JA\&LB$ strategy. Specifically, considering a rectangular HetECN mission area with the size of 2 km × 2 km, unless otherwise specified, 400 GUs are randomly distributed within this area. All ABSs take off from the position of EV. Assuming the scenario of disaster rescue action in urban area, according to [51], the low urban environments are $a = 9.61$ and $b = 0.16$, $\eta_{\text{LoS}} = 1$, $\eta_{\text{NLoS}} = 20$. The carrier frequency is $f_c = 28$ GHz. During initialization, we set the maximum path loss to 95 dB, the optimal initial altitude of ABS is calculated to be $h_{ini} = 340$ m, and the maximum ABS coverage radius is set to $R = 270$ m. Based on network traffic demands and considering capacity margin in emergency situations, the number of ABS provide service is 10.

Fig. 6 shows the trajectory of 3 ABSs under combined virtual forces in HetECN. The cylinder represents the obstacle after calculating the circumcircle, the hollow circle represents the position of AMJ, and the shape of hills are 3D thermal map of the GU hotspot distribution when GU = 200. To better illustrate the ABS trajectory, we also present its top view in Fig. 6(b). It is not difficult to see that using the proposed JAA algorithm, ABS can accurately avoid obstacles and jamming ABS during navigation, and also achieve interference avoidance between service ABS, ultimately finding the best hovering position.

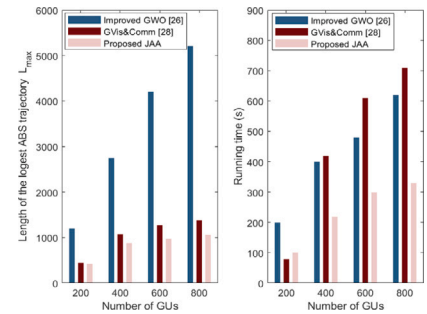



**Fig. 7.** The longest path length and algorithm running time under different network scales.

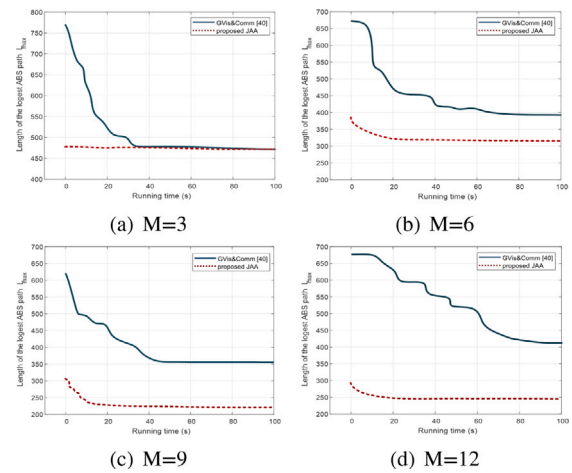



**Fig. 8.** Convergence analysis of JAA and improved GWO under different ABS quantities when N = 500.

Fig. 7 indicate that the length and running time of the longest trajectory among all ABSs increase with the increase of network size. Among the three trajectory optimization methods, the improved GWO algorithm [26] under complex confrontation environment has a fast $L_{max}$ increment speed due to the lack of practical and feasible methods to adjust the structure of the longest trajectory. Regardless of how the network size increases, both GVis&Comm [28] and JAA exhibit satisfactory performance with $L_{max} \leqslant 1000$ m. Although GVis&Comm and JAA have similar performance in terms of the longest trajectory length, GVis&Comm exhibit an exponential increase in runtime as the network size expands. So when the network size exceeds GU = 600, GVis&Comm becomes the most time-consuming algorithm. Our proposed JAA algorithm simultaneously reduces the length and planning time of the trajectory under the condition of a surge in network traffic demand, making it a feasible and effective trajectory optimization method.

In Fig. 8, when the number of GU is 500, the convergence speed of JAA algorithm and improved GWO is compared by changing the number of ABS. From Fig. 8(a), it can be observed that when the number of ABS is small, JAA and improved GWO algorithm can quickly converge to similar $L_{max}$ values. However, as $M$ increases, the value of $L_{max}$ separates between JAA and improved GWO. When there is a large number of ABS, the convergence speed of improved GWO tends to flatten, while JAA can still maintain a fast convergence speed within

**Table 2**
Capacity of each ABS.

| Index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| SIR-based [47] | 0.0461 | 0.0703 | 0.0367 | 0.0548 | 0.0498 | 0.0583 | 0.0332 | 0.0672 | 0.0723 | 0.0356 |
| Distance-based [40] | 0.0612 | 0.0432 | 0.0363 | 0.0617 | 0.0493 | 0.0435 | 0.0381 | 0.0498 | 0.0641 | 0.0275 |
| MAGame-LB | 0.0623 | 0.0574 | 0.0528 | 0.0637 | 0.0525 | 0.0637 | 0.0426 | 0.0479 | 0.0583 | 0.0428 |

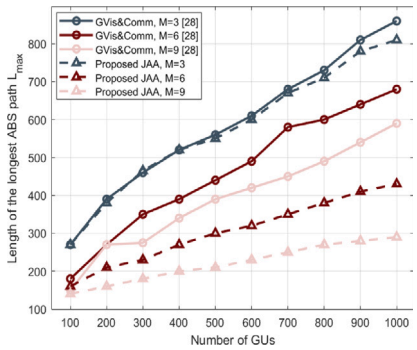



**Fig. 9.** The length of longest path $L_{max}$ of ABS under different network scales.

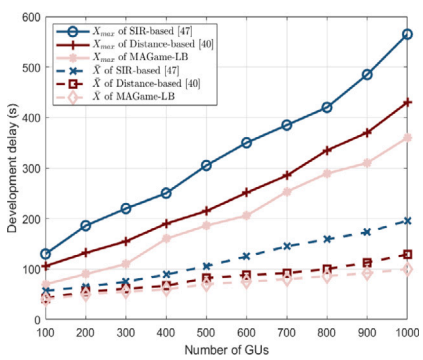


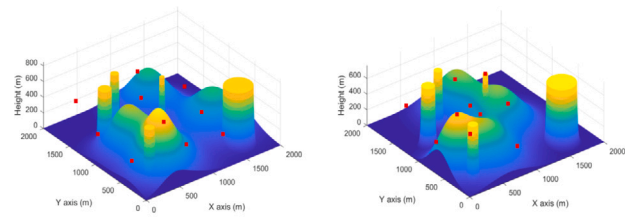

**Fig. 10.** Comparison between maximum deployment delay and average deployment delay.

10 s (see Figs. 8(b)–8(d)). To demonstrate the effectiveness of our proposed JAA algorithm, we further measure its performance through changes in network scale.

Fig. 9 shows the comparison results by changing the number of GU (100 to 1000 GUs, with an interval of 100). Obviously, when the number of ABSs is 3, JAA and GVis&Comm algorithm can maintain almost equivalent performance until the network scale exceeds 700. In contrast, as the number of GUs or network scale increases, the gap between JAA and GVis&Comm continues to widen. When M = 9 and the number of GU = 1000, The longest path length is reduced by 42%, which means significantly decrease the network deployment latency.

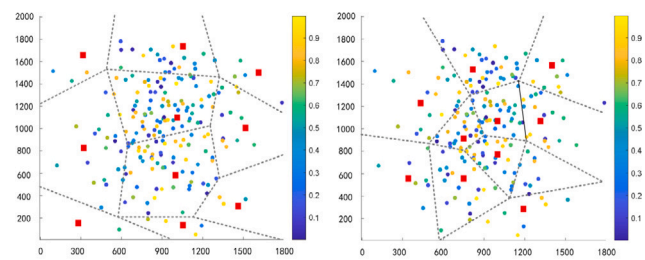
Fig. 10 shows a comparison of deployment delays between MAGame-LB and GVis&Comm. It is not difficult to see that as the network size increases, both the maximum deployment delay and the average deployment delay increase. When the network size is small ($\mathcal{N} \leq 300$), compared with GVis&Comm, MAGame-LB can achieve similar communication latency. This indicates that when the network size is small, the two algorithms have the same performance in communication delay. However, when the network size is large enough ($\mathcal{N} \geq 400$), the deployment delay of GVis&Comm increases by 23% and 47% respectively compared to GVis&Comm. Therefore, MAGame-LB is superior to GVis&Comm in terms of deployment latency and is suitable for the high requirements of network deployment timeliness in HetECN.

For network load balancing performance evaluation, we compared the GU-ABS association scheme based on our proposed MAGame-LB algorithm with SIR-based scheme (our previous work that GU-ABS




(a) Distribute-based HetECN (b) Demand-based HetECN

**Fig. 11.** Comparison of ABS hover position based on GU demand and GU density.



(a) Distribute-based (b) Demand-based

**Fig. 12.** Vertical view of ABS hover position based on GU demand and GU density.

**Table 3**
Performance of ABS altitude adjustment method.

| Index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| After fine-tuning | 268 | 279 | 341 | 324 | 359 | 246 | 283 | 327 | 316 | 261 |

**Table 4**
Performance of MAGame-LB algorithm.

| | SIR-based [47] | Distance-based [40] | MAGame-LB |
|---|---|---|---|
| $\mathbb{U}$ | 4.3% | 4.1% | 2.8% |
| $\mathbb{R}$ | 0.0451 | 0.0457 | 0.0462 |
| $\mathbb{L}$ | 0.0021 | 0.0019 | 0.0018 |
| $\mathbb{A}$ | 69.8611 | 63.4952 | 56.4865 |

association is determined by the optimal channel conditions) [47] and distance-based algorithm [40]. As shown in Figs. 11 and 12, it can be seen that the position of ABSs derived from our proposed MAGame-LB algorithm is at the center of the maximum local density region. Table 2 shows the actual load capacity of each ABS. It can be noticed that the maximum traffic load of ABS in SIR-based scheme is 0.0723, which is much higher than 0.0637 in MAGame-LB.

In Table 4, we compare the results of the three GU-ABS association schemes through four evaluation metrics mentioned earlier. It is not difficult to see that the proportion of GUs who cannot be served in MAGame-LB is 2.8%, and the average load capacity of ABSs is 0.0462, both showing the best ABS service performance. In addition, the load balancing index between ABSs is 0.0018, and the regional load balancing index between ABS is 56.4865, both much lower than that of SIR-based and distribution-based algorithm. So we can conclude that the proposed MAGame-LB algorithm can significantly improve the load balancing ability of ABS in HetECN, thereby serving more GUs. In addition, our game algorithm requires approximately 70 iterations to

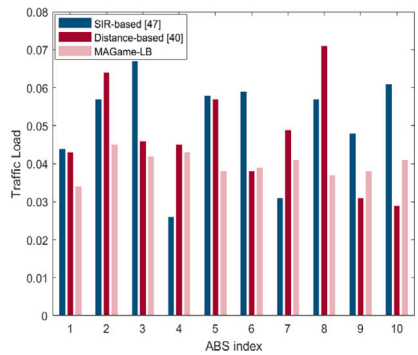



**Fig. 13.** Load balancing capability evaluation.

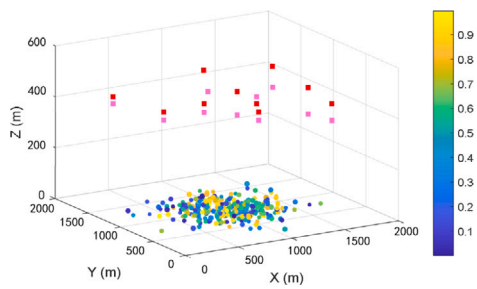



**Fig. 14.** ABS altitude fine-tuning, GU = 300.

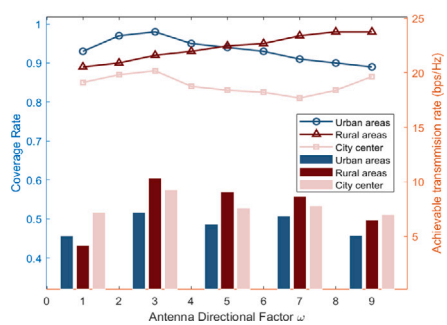



**Fig. 15.** The coverage rate and achievable transmission rate of HetECN in different scenarios.

converge to local optima. In other words, the proposed algorithm has low complexity and therefore has practical value in HetECN.

As shown in Fig. 13, compared with other two algorithms, MAGame-LB significantly reduces the difference in traffic load between ABSs, as MAGame-LB attempts to balance the traffic load between ABS by associating GU with appropriate ABS in each iteration. In contrast, SIR-based scheme associates GU with ABS with optimal channel conditions without considering the traffic load balancing of these ABSs, resulting in network imbalanced traffic load. In distance-based algorithm, GU-ABS association is determined by the coverage area of ABS according to ABS coverage radius, without considering the differences in GU traffic requirements in HetECN, also lead to imbalanced traffic load between ABSs.

As shown in Table 3 and Fig. 14, the altitudes of 10 ABSs after fine-tuning are displayed. It is not difficult to see that there are 8 ABSs hovering below the default altitude of 340 m, which makes network deployment resources more efficient. Therefore, the ABS altitude fine-tuning strategy to reduce ABS power consumption is helpful and necessary for cater to the limited resources and network characteristics of HetECN.

Fig. 15 demonstrates the relationship between coverage rate, achievable transmission rate and $\omega$ in different HetECN scenarios, with the environmental parameters (a,b) for rural, urban, and city center settings being (5.30,0.45), (9.61,0.16), and (23.2,0.07) respectively. It is observed that the network coverage rate in urban areas is relatively the highest. As $\omega$ increases, the coverage rate in city center first increases and then decreases due to the blocking effect of high-rise buildings on mmWave signals, resulting in a maximum difference of 9.8% between the highest and lowest coverage rate. In rural areas, where GU demands are smaller, mostly consisting of sensor devices and with fewer obstacles, the network coverage probability can increase by 12% when antenna directional factor $\omega$ expanded from 2 to 8. In addition, the city center has a lower achievable transmission rate due to the shadow effect of obstacles and various external interference.

## 6. Conclusion

In this article, we designed an EV based mmWave UAV-assisted HetECN model. Specifically, the jamming avoidance ABS trajectory design and load balancing GU-ABS association problem was studied in HetECN. Firstly, an improved APF method was introduced to deal with various obstacles and interference ABS encountered, following the direction of the combined force to find optimal path and hover position of ABSs. Then, we studied the GU-ABS association based on load balancing, which attempts to minimize the maximum traffic demand of each ABS and balance the traffic load between them as much as possible. Finally, utilizing ABS altitude fine-tuning was used to further reduce the overall power consumption of HetECN. Simulation results show that compared with other schemes, our scheme can serve more GUs, save network deployment time and cost, and make the traffic load between UAVs more balanced. In future work, optimizing UAV load balancing in dynamic GU scenarios and focus on GUs' motion mode prediction is to be considered. What is more, the current situation of dynamic interference deserves further exploration.

## CRediT authorship contribution statement

**Xingchen Wei:** Writing – review & editing, Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Laixian Peng:** Writing – review & editing, Validation, Supervision, Resources, Project administration. **Renhui Xu:** Validation, Software, Resources, Methodology, Data curation. **Hai Wang:** Validation, Investigation, Funding acquisition.

## Declaration of competing interest

(1) All authors disclosed no relevant relationships.

(2) The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

## Data availability

The authors do not have permission to share data.

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